GaussFM: Compact Foundation-Feature Gaussian Memory for Open-Vocabulary 3D Scene Understanding

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***Abstract*—Dense vision foundation models provide strong 2D features, but deployed 3D scenes still lack a compact scene memory that can carry those features across novel views, Gaussian primitives, and point-query protocols. We present GaussFM, a compact foundation-feature Gaussian memory for 3D scenes. Rather than storing raw 1280-D RADIO features on every Gaussian or training a scene-specific classifier, GaussFM stores low-dimensional per-Gaussian latent codes, augments them with spatial context and reliability cues, and reconstructs RADIO-compatible scene features on demand. Training combines dense rendered RADIO reconstruction with sparse Multiview Primitive Registration, which compresses registered multiview semantic evidence back into the same compact Gaussian memory. The resulting representation supports rendered 2D querying, direct primitive selection, and direct point querying without deploying a multiview feature cache.**

We evaluate GaussFM under three protocols. On LERF-OVS rendered-view open-vocabulary query, GaussFM reaches 64.98 mIoU and 82.68 Acc. Under the same LERF-OVS direct 3D query-select-render protocol, compact primitive scoring with support calibration reaches 54.36 mIoU and 80.84 Acc without reading a multiview-registration cache or invoking an official RGB SAM decoder at inference. On the VALA-aligned ScanNet-8 direct point-query protocol, the compact field reaches 36.55, 42.78, and 57.85 mIoU on the 19-, 15-, and 10-class splits. Same-evaluator frame-wise foundation-feature comparisons and component ablations show that multiview reconstruction, sparse primitive anchoring, and support-calibrated 3D selection drive the gains. Small fragmented objects remain the primary failure mode.

***Index Terms*—Open-vocabulary 3D scene understanding, 3D Gaussian Splatting, vision foundation models, feature distillation, scene representation.**

I. Introduction

Vision foundation models expose dense features that are useful for recognition, localization, segmentation, and geometric reasoning. However, most 3D scene representations still optimize either RGB appearance or task-specific labels. In open-vocabulary 3D understanding, this creates a gap: the strongest 2D feature extractors are available at training images, but a deployed 3D scene must answer queries from novel views, select objects directly in 3D, and transfer to new point-query protocols without storing a dense RADIO feature on every primitive.

GaussFM targets this gap by reconstructing foundation-model features inside a 3D Gaussian scene. Our goal is not to train a new per-scene language classifier. Instead, we ask whether one compact scene memory can support multiple query modes: dense novel-view feature maps for rendered open-vocabulary localization, direct primitive scores for 3D object selection, and point-level features for semantic querying. This feature-reconstruction framing differs from methods that attach only a final language code to a radiance field or optimize a single downstream score.

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The technical challenge is that RADIO features are high-dimensional, dense, and expensive to store directly on every Gaussian. We therefore learn a compact Contextual Gaussian Feature Field, reconstruct foundation-compatible features through a Foundation-Space Reconstruction module, and calibrate them in view space before comparison to frozen RADIO targets. We also use Frozen-Head Geometry Consistency as a geometry-aware regularizer: a frozen downstream head guides the feature field without turning the model into a task-specific predictor.

We evaluate this formulation with a protocol boundary that mirrors recent open-vocabulary Gaussian-field papers. LERF-OVS rendered-view grounding tests whether the field can produce dense novel-view features; LERF direct 3D object selection tests whether text queries can select Gaussian primitives before any rendered-mask evaluation; and VALA-aligned ScanNet-8 direct point querying tests whether the same compact representation remains usable outside the LERF object-query setting. Baseline comparisons in the main LERF and ScanNet tables follow the same reproduced protocols as GaussFM, while historical provenance audits are kept in the appendix.

a) Contributions.: We make three contributions:

- We introduce a compact reconstructive RADIO Gaussian feature field: low-dimensional per-Gaussian latent codes, spatial context, reliability modeling, and global decoders replace explicit high-dimensional feature storage while preserving foundation-feature usability.
- We combine dense rendered RADIO reconstruction with sparse Multiview Primitive Registration and support calibration, so the same compact memory supports rendered 2D querying, direct 3D primitive selection, and ScanNet point-level querying without a deployed multiview feature cache.
- We provide a protocol-separated evaluation covering LERF-OVS 2D and 3D open-vocabulary query, VALA-aligned ScanNet point query, frame-wise RADIO versus reconstructed scene features, storage/latency, qualitative analysis, and component ablations.

II. Related Work

a) Language fields and open-vocabulary 3D understanding.: LERF embeds language-aligned features in neural radiance fields and enables open-vocabulary querying in 3D scenes [1]. LangSplat and Language Embedded 3D Gaussians adapt this direction to Gaussian scene representations [2], [3]. OpenGaussian formalizes a stronger primitive-level protocol in which text selects 3D Gaussians that are then rendered only for mask evaluation [4]. Recent methods push this direction further: Dr. Splat registers language-aligned features directly to dominant Gaussians along pixel rays [5], while InstanceGaussian and OpenSplat3D move toward instance-level Gaussian grouping and open-vocabulary instance segmentation [6], [7]. OpenInsGaussian further studies context-aware cross-view fusion for instance Gaussian segmentation [8], while very recent Ilov3Splat uses view-consistent feature fields and two-stage 3D clustering for instance-level open-vocabulary querying [9]. SuperGSeg clusters Gaussians into structured super-primitives for hierarchical understanding [10]. These methods demonstrate that Gaussian scenes can carry language-aligned evidence, but they mostly store or aggregate final semantic embeddings. They do not directly answer whether a 3DGS scene can be a compact reconstructive memory for a high-dimensional foundation feature while retaining rendered-view, primitive-level, point-level, and frozen-head downstream usability. GaussFM targets this gap by reconstructing RADIO-compatible features from low-dimensional Gaussian codes and compressing registered multiview evidence back into the same primitive-level memory.

b) Gaussian scene representations.: 3D Gaussian Splatting provides an explicit and efficient scene representation for radiance-field rendering [11]. GaussFM keeps this geometry backbone and learns a feature field on top of it, separating geometric visibility from foundation-feature reconstruction.

c) Agglomerative vision foundation models.: RADIO-style models distill multiple visual capabilities into a single dense vision backbone [12], [13]. GaussFM uses C-RADIOv4-H as a frozen feature source and studies whether its dense features can be reconstructed from 3D scenes at novel views.

d) Foundation-model regularization in 3DGS.: FMGS embeds 2D foundation-model features into Gaussian scenes and uses DINO-style spatial structure to improve feature consistency [14]. SAM-based Gaussian methods such as SAGA distill 2D segmentation priors into 3D Gaussian features for promptable object segmentation [15]. Our adaptor regularizers follow the same principle, but use the official C-RADIOv4-H adaptor heads so that the reconstructed feature remains a single RADIO-compatible representation at inference time.

III. Method

A. Problem Setup

Given posed RGB training images and a pretrained 3D Gaussian scene, our goal is to learn a compact feature field that is compatible with frozen RADIO features and can be queried through both rendered views and 3D locations. Let I_v be an image at view v and $T(\cdot)$ be the frozen RADIO encoder. The rendered RADIO target is:

$$F_v^\star = T(I_v) \in \mathbb{R}^{H \times W \times 1280}. \tag{1}$$

We learn a Gaussian feature field that renders $\hat{F}_v$ and minimizes feature reconstruction losses against $F_v^\star$. The same compact field can also be decoded at Gaussian centers or query points, which makes direct primitive and point-level querying part of the representation rather than separate task-specific classifiers. The RGB/geometry 3DGS scene is reconstructed before feature-memory learning and kept fixed during the experiments reported here. GaussFM does not introduce an RGB reconstruction loss or claim improved radiance-field appearance; its trainable state is the compact foundation-feature memory and its global reconstruction and calibration heads.

B. Contextual Gaussian Feature Field

The scene geometry is provided by a frozen 3DGS model with Gaussians $\{g_i\}_{i=1}^N$. GaussFM attaches compact learnable latent features to the Gaussians and combines two paths:

1) a fine path that splats per-Gaussian latent features into the image plane;
2) a coarse path that queries a spatial feature branch from reconstructed 3D positions.

The two paths are fused by decoupled geometry and semantic heads into a compact feature map C_v. In the unified multi-head field, the same fused state also predicts a feature-quality logit q_v and a visibility logit u_v:

$$(C_v, q_v, u_v) = H_\phi(Z_v^{fine}, Z_v^{coarse}, d_v, \alpha_v). \tag{2}$$

The quality head estimates whether a rendered compact feature will reconstruct the RADIO target reliably, while the visibility head estimates whether the rendered evidence is supported by visible Gaussian mass. Both heads support rendered-view feature reconstruction and direct Gaussian/point query calibration. A Foundation-Space Reconstruction module maps this compact representation back to RADIO feature space:

$$\hat{F}_v = D_\theta(C_v). \tag{3}$$

A View-Conditioned Feature Calibration module then improves local alignment using rendered guidance signals such as visibility, alpha, and geometric depth. All downstream adaptor scores are computed after this RADIO-space reconstruction. The compact latent itself is therefore not a stored SigLIP, DINO, or SAM embedding; those spaces constrain or probe decoded RADIO-compatible features.

C. Training Objective

The base feature loss combines cosine and reconstruction terms:

$$\mathcal{L}_{feat} = 1 - \frac{\langle \hat{F}_v, F_v^\star \rangle}{\|\hat{F}_v\|_2 \|F_v^\star\|_2} + \lambda_{rec} \|\hat{F}_v - F_v^\star\|_1. \tag{4}$$

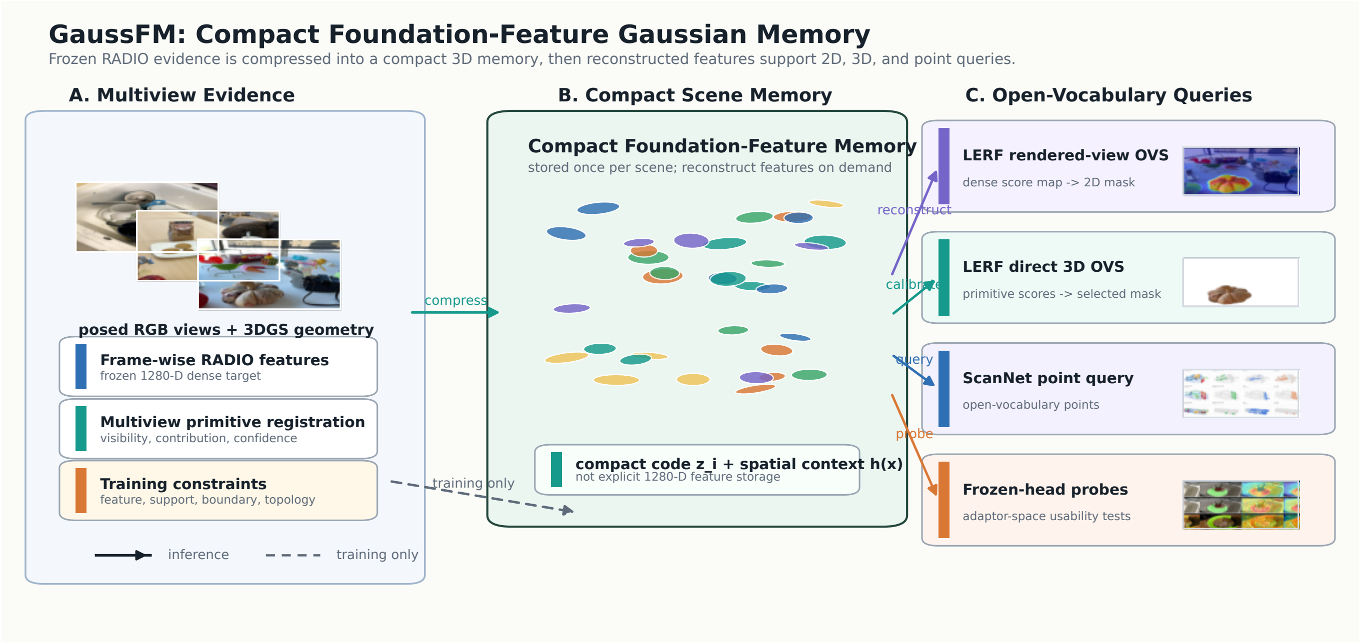

Fig. 1. GaussFM framework. A pretrained 3DGS scene supplies geometry, while frozen RADIO supplies the dense foundation feature to be reconstructed. The method stores compact per-Gaussian latent codes with spatial context and quality/visibility cues, then reconstructs RADIO-compatible scene features by global decoders. SigLIP2, DINOv3, and SAM3 adaptor spaces are used as frozen-head constraints or probes of the reconstructed RADIO feature, not as separate raw feature memories. Multiview Primitive Registration compresses registered semantic evidence into the same compact memory during training. At inference, the memory supports rendered 2D query, direct Gaussian selection, and point-level query without a multiview-registration feature cache or official RGB SAM decoder.

For the FGC route, we add a frozen depth-head consistency loss. Let h_d be a frozen depth head and d_v^{geo} be the geometric depth rendered from the 3DGS backbone. We optimize:

$$\mathcal{L} = \mathcal{L}_{feat} + \lambda_{fgc}\mathcal{L}_{si}(h_d(\hat{F}_v), d_v^{geo}), \tag{5}$$

where $\mathcal{L}_{si}$ is a scale-invariant depth loss. The head is frozen; the feature field must adapt so that its rendered features remain useful to the head.

The unified quality heads add two label-free auxiliary objectives:

$$y_v^q = \frac{1 + \cos(\hat{F}_v, F_v^\star)}{2}, \qquad y_v^u = \alpha_v, \tag{6}$$

where gradients are stopped through the targets. We train q_v and u_v with binary cross-entropy against y_v^q and y_v^u respectively. These losses do not introduce task labels; they expose feature reliability and visibility as first-class field outputs that can be reused by text grounding, direct primitive scoring, and failure analysis.

D. Dense Reconstruction and Adaptor-Space Regularization

C-RADIOv4-H is the primary foundation feature learned by the compact Gaussian memory. The stored latent codes are not DINO, SAM, or SigLIP features; they are decoded into RADIO-compatible features and then tested or regularized through frozen adaptor spaces. We therefore use the adaptor heads as constraints on the reconstructed RADIO feature, not as separate scene memories. For an adaptor A_a, the generic dense consistency term is:

$$\mathcal{L}_a = 1 - \cos(A_a(\hat{F}_v), A_a(F_v^\star)). \tag{7}$$

Dense structural adaptors are used where their supervision matches the rendered feature-map setting. For `dino_v3`, we also match the token self-similarity structure rather than only the local feature value:

$$\mathcal{L}_{rel} = \|DD^\top - D^\star D^{\star\top}\|_2^2. \tag{8}$$

This transfers DINO-like boundary and correspondence structure into the rendered RADIO feature field, similar in spirit to DINO regularization in FMGS. Cross-view DINO affinity and dense SigLIP text-heatmap terms are evaluated as ablations, but they are not the core rendered reconstruction objective. The dense/sparse SigLIP study shows that dense text supervision is less important for rendered feature quality than sparse primitive anchoring is for direct 3D querying. This division is intentional. DINO and SAM adaptor signals encode dense neighborhood topology, region continuity, and boundary structure; applying them only at isolated primitive centers would discard the very spatial relations that make them useful. SigLIP2 summary features, by contrast, are designed as global/text-aligned semantic descriptors and are therefore better suited as sparse semantic anchors for visible primitives. The sparse branch is thus a semantic-usability regularizer on decoded RADIO features, not a task-specific replacement for dense RADIO reconstruction.

For `sam3`, the dense training branch uses a mask-free soft-region loss. Frame-wise SAM3-adaptor tokens define deterministic anchor prototypes; each pixel is softly assigned to anchors by adaptor-space similarity, and the rendered feature

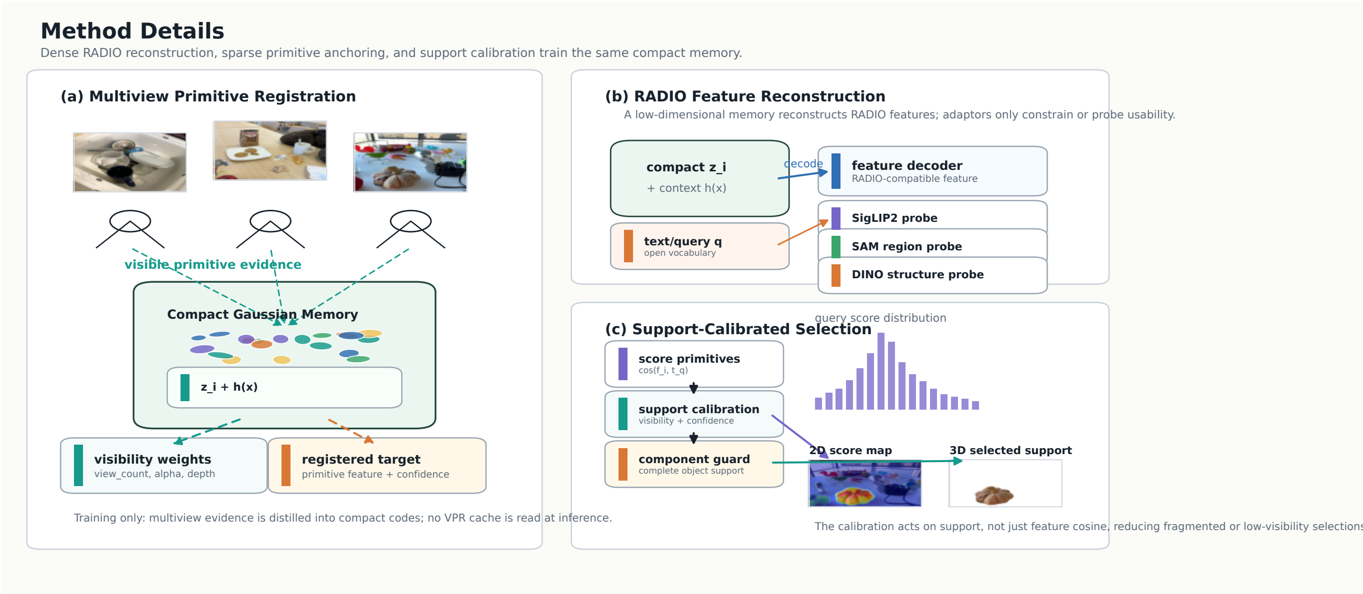

Fig. 2. Method details. Dense rendered RADIO reconstruction teaches the compact Gaussian memory to recover foundation-compatible feature maps; sparse Multiview Primitive Registration anchors visible primitives in a text-summary space during training; support calibration turns primitive scores into stable object support at query time. DINO/SAM/SigLIP adaptor spaces constrain or probe the reconstructed RADIO feature rather than becoming separate stored memories.

is trained to match frame-wise region prototypes:

$$\mathcal{L}_{reg} = \sum_k \|\mu_k(A_{\mathrm{sam3}}(\hat{F}_v)) - \mu_k(A_{\mathrm{sam3}}(F_v^\star))\|_2^2. \tag{9}$$

This approximates SAM-style region supervision without requiring external SAM mask caches, while keeping the inference-time representation unchanged. A mask-logit variant is retained in the appendix as a diagnostic. All SAM losses remain adaptor-space constraints rather than official frozen SAM3 mask-decoder supervision. The official SAM3 decoder is used only in separate diagnostic or pseudo-mask paths; the main paper claims are based on reconstructed RADIO features and adaptor-space probes rather than an external RGB SAM decoder.

E. Warm-Start Protocol

Directly applying frozen-head losses early in training can conflict with feature reconstruction. The mainline therefore uses a warm-start schedule: first train a feature-only field, then initialize the FGC model from that checkpoint and continue training with the frozen geometry-head loss active. This makes FGC a late geometry-aware refinement signal instead of an early competing objective.

F. Sparse Primitive Anchoring by Multiview Registration

Multiview Primitive Registration is a training and analysis bridge, not a deployed feature cache. It renders decoded RADIO-space features from posed views, applies view-conditioned calibration and the frozen SigLIP2 summary head, and registers the resulting text-aligned summaries back to visible Gaussian centers with depth/alpha checks. This produces sparse primitive-level semantic targets from multiview evidence while keeping the stored representation as a compact RADIO-reconstructive memory.

To make the direct compact field itself more usable, we train a label-free multiview-registration-to-field consistency variant. Let $\bar{e}_i^{\mathrm{mpr}}$ be the registered normalized summary target for Gaussian i, and let n_i be the number of posed views that registered valid evidence to Gaussian i. The transfer objective uses a label-free confidence weight $w_i = \log(1+n_i)/\mathrm{mean}_j \log(1+n_j)$ over valid registered primitives. We fine-tune the compact field and a lightweight summary adaptor with

$$\mathcal{L}_{v2f} = 1 - \cos(A_{\mathrm{SigLIP2}}(D_\theta(c_i)), \mathrm{sg}[\bar{e}_i^{\mathrm{mpr}}]) + \lambda\, \mathcal{L}_{text},$$

where $\mathcal{L}_{text}$ matches the registered multiview scene-query distribution and uses only the LERF text categories, not object masks; all loss terms are averaged with w_i. Importantly, the supervision path still decodes c_i to RADIO space before the SigLIP2 summary head is applied. The compact latent is never directly stored or optimized as a standalone SigLIP embedding. At inference, the main compact memory does not read the registration targets. For the Dr. Splat-inspired audit, we also implement rasterizer-level Gaussian–pixel hit registration variants: all footprint contributions, per-pixel dominant hits, and per-Gaussian top-contribution pixels. These variants capture real rasterization intersections instead of only projected centers, but they are treated as negative controls because they reduce the LERF direct-3D metrics under the fixed protocol.

G. Query Modes and Support Calibration

At evaluation time, GaussFM renders a feature map for each novel view. For a text query q, we compute a text embedding

e_q and score each pixel by feature-text similarity with a scene-category softmax temperature τ. The predicted localization point is the heatmap maximum. LERF-OVS reports localization accuracy and mIoU over query heatmaps.

For direct 3D querying, the base Gaussian-center query decodes pre-refiner compact features at Gaussian centers:

$$\hat{f}_i = D_\theta(c_i), \qquad s_i(q) = \cos(A_{\mathrm{SigLIP2}}(\hat{f}_i), e_q),$$

where A_{SigLIP2} is the frozen text-aligned summary head. This uses the same compact Gaussian memory as the ScanNet point-query experiments. For LERF primitive selection, we use a frozen SigLIP2 prompt ensemble and support-calibrated primitive selection: selected Gaussians are rendered to a coarse mask, label-free color-edge support calibration refines local boundaries, and a component guard keeps the largest component only when it dominates the refined support; otherwise it preserves multi-component support only above a fixed support-area floor. This is a label-free support rule. It uses the evaluation RGB image only through a classical color-edge support refinement, not through a learned feature extractor or official SAM decoder, and uses LERF masks only for the final query-select-render evaluation.

IV. Experiments

A. Evaluation Protocol and Provenance

All GaussFM numbers are tied to fixed evaluation outputs and a provenance manifest. The rendered-view and direct-3D LERF runs use the same four annotated LERF-OVS scenes; the ScanNet study uses the VALA-aligned eight-scene subset and the class splits described below. The main quantitative baseline rows are reported under the same reproduced evaluation protocols as GaussFM; historical source/provenance notes are retained separately for auditability.

B. LERF-OVS Main Result

Table I provides the same-protocol LERF-OVS 2D and 3D open-vocabulary query comparison. The 2D rows evaluate rendered-view query masks, while the 3D rows query Gaussian primitives first and render selected support only for mask evaluation. The GaussFM row gives the highest mean mIoU in 2D and the highest mean mIoU/Acc in direct 3D. For 2D query, the compact RADIO feature field renders a dense RADIO-compatible feature map; the resulting query heatmap is converted to a mask by the fixed protocol used for the reproduced baseline comparison.

Supplementary calibration isolates the rendered-mask boundary head from feature quality. Under a controlled PCC-only mask conversion, raising the label-free peak-relative threshold from 0.50 to 0.60 tightens masks and improves macro mIoU from 0.494 to 0.524, while peak-connected-component filtering lifts the same four-scene mask conversion to 0.5707. This controlled number is not the final 2D benchmark row. The final reproduced 2D OVS row in Table I further uses the feature-only SAM3-adaptor boundary head and reaches 64.98 mIoU. The feature-only SAM3-adaptor boundary head is therefore used only after the peak component has supplied a stable coarse support, turning it into a boundary refinement step rather than a new localization mechanism.

C. LERF Direct 3D Object Selection

Table II evaluates the primitive-level bridge to 2024–2025 open-vocabulary 3DGS work. We follow the OpenGaussian-style query-select-render protocol: text similarity is computed on 3D Gaussian primitives, selected primitives are rendered to official LERF-OVS annotated views, and metrics are mIoU and Acc@0.25. This protocol is not the same as the rendered-view heatmap evaluation in Table I; the query is performed in 3D and rendering is used only to compare selected primitives with 2D masks.

Table III separates two kinds of evidence. The controlled rows isolate prompt and component-support effects: a strict no-prompt/no-RGB compact one-map query reaches 0.449 macro mIoU and 0.672 Acc@0.25; adding only the frozen prompt ensemble gives 0.457/0.685; adding GT-free color-edge component support gives 0.500/0.705. The final row is the complete fixed-protocol compact-memory result from Table I. It should not be read as the next single-module step in the controlled factorial; the controlled rows isolate partial mechanisms, while the last row records the deployed selector used for the benchmark ranking. In all cases the compact primitive query does not read a multiview-registration feature cache or use an official SAM3 mask decoder.

The frozen official SAM3 box-prompt result is reported separately as a boundary diagnostic after compact 3D selection. It reaches 0.570 macro mIoU and 0.684 Acc@0.25, but SAM3 is invoked only after primitives have already been scored and rendered; its candidate mask is selected by overlap with the rendered prediction rather than ground truth. This result is useful for measuring the remaining boundary gap, but the core direct-field claim is carried by the compact memory and its support-calibrated primitive selection. Waldo Kitchen still shows that instance grouping can help fragmented objects, so we treat these results as evidence for direct-field usability rather than as a universal primitive-level ranking claim.

Supplementary protocol audits verify that Multiview Primitive Registration uses 128 posed rendered views, visibility checks, and no LERF masks or query labels for registration. They also record negative or auxiliary variants such as fixed-ratio selectors, best-by-scene thresholds, contribution-weighted raster registration, proposal/OPR component selection, and confidence selectors. The main conclusion is that feature registration improves primitive evidence, but stable object selection requires support-calibrated primitive selection; query-level audits identify Waldo Kitchen as a coverage, ambiguity, and boundary-fragmentation bottleneck.

D. Rendered Features vs. Original RADIO Features

Table IV supports the inference-time feature-memory claim: the rendered feature field is not merely a lossy copy of frame-wise RADIO. It can sharpen the task-relevant scene response after multiview reconstruction: under the same calibrated mask conversion it improves both localization and macro mIoU over frame-wise RADIO RGB features, even though the original RADIO RGB feature remains the direct 2D training target.

Two controls are retained in the supplementary material. A nearest-view RADIO cache without geometric warping

TABLE I
QUANTITATIVE EVALUATION OF 2D AND 3D OPEN-VOCABULARY QUERY ON LERF-OVS. ALL BASELINE ROWS ARE REPRODUCED UNDER THE SAME PROTOCOL AS GAUSSFM. WE REPORT MIOU AND ACC IN PERCENT; BEST RESULTS ARE IN **BOLD**.

Eval.	Method	Mean		Figurines		Teatime		Ramen		Waldo Kitchen	
		mIoU↑	Acc↑	mIoU	Acc	mIoU	Acc	mIoU	Acc	mIoU	Acc
2D	LangSplat	51.40	**84.30**	44.70	80.40	65.10	88.10	51.20	73.20	44.50	**95.50**
	GAGS	54.12	81.66	53.59	78.57	60.29	88.14	46.81	69.01	55.80	90.91
	OccamLGS	61.30	82.50	58.60	80.40	70.20	93.20	51.00	74.70	65.30	81.80
	GOI	42.00	59.20	23.90	44.60	55.80	67.80	33.70	56.30	54.50	68.20
	GALA	55.49	73.43	59.35	82.14	**76.73**	88.14	35.13	50.70	50.75	72.73
	LangSplatV2	59.90	84.10	56.40	82.10	72.20	93.20	51.80	74.70	59.10	**95.50**
	GaussFM	**64.98**	82.68	**64.29**	**92.86**	76.09	**93.22**	**53.78**	62.83	**65.76**	81.82
3D	OpenGaussian	38.36	51.43	39.29	55.36	60.44	76.27	31.01	42.25	22.70	31.82
	SuperGSeg	35.94	52.02	43.68	60.71	55.31	77.97	18.07	23.94	26.71	45.45
	OccamLGS	47.22	74.84	52.90	78.57	61.02	**93.22**	32.01	54.92	42.95	72.72
	Dr. Splat	43.29	64.30	54.42	80.36	57.35	77.97	24.33	35.21	37.05	63.64
	GALA	36.71	59.71	45.25	69.64	53.27	84.75	17.08	25.35	31.22	59.09
	LangSplatV2	35.87	55.80	45.15	67.86	49.30	79.66	19.01	21.13	30.00	54.55
	GaussFM	**54.36**	**80.84**	**59.36**	**92.86**	**71.04**	89.83	**38.43**	**63.38**	**48.61**	**77.27**

TABLE II
LERF-OVS DIRECT 3D OPEN-VOCABULARY QUERY UNDER THE SAME REPRODUCED QUERY-SELECT-RENDER PROTOCOL. WE REPORT MIOU AND ACC IN PERCENT.

Method	mIoU↑	Acc↑
OpenGaussian	38.36	51.43
SuperGSeg	35.94	52.02
OccamLGS	47.22	74.84
Dr. Splat	43.29	64.30
GALA	36.71	59.71
LangSplatV2	35.87	55.80
GaussFM	**54.36**	**80.84**

TABLE III
COMPACT DIRECT3D READOUT ABLATION. ALL ROWS USE COMPACT GAUSSIAN-CENTER PRIMITIVE SCORES AND THE OPENGAUSSIAN-STYLE QUERY-SELECT-RENDER EVALUATION. THE PURE ONE-MAP ROW IS THE STRICT NO-VPR/NO-RGB/NO-SAM READOUT; GUARDED ROWS ADD GT-FREE COMPONENT SUPPORT POLICIES BUT STILL AVOID A VPR CACHE AND OFFICIAL RGB SAM DECODER.

Readout	VPR	SAM3	RGB	Prompt	mIoU	Acc@0.25
Compact single prompt, pure one-map	no	no	no	no	0.449	0.672
Compact direct, previous promoted baseline	no	no	yes	no	0.484	0.643
Compact prompt ensemble, pure one-map	no	no	no	yes	0.457	0.685
Compact prompt ensemble + RGB component guard	no	no	yes	yes	0.500	0.705
Compact prompt ensemble + RGB/score-component guard	no	no	yes	yes	0.501	0.704

reaches only 0.2722 LocAcc / 0.1545 mIoU, and a raw per-Gaussian 1280-D foundation-feature memory reaches 0.5642 / 0.3182 while using 1039.7 MiB mean feature storage. Both trail compact GaussFM, supporting the claim that learned multiview feature reconstruction is more useful than either a 2D cache or sparse high-dimensional foundation-feature attachment. We additionally consolidate the frame-wise RADIO versus GaussFM evidence across SigLIP2 text grounding and frozen SAM3/DINOv3 task probes in a supplementary table: rendered GaussFM features win all 6 selected primary downstream metrics. This summary is intentionally phrased as selected downstream feature usability, because several secondary LocAcc/HitRate probes remain stronger for the frame-wise RADIO features under the same probe.

TABLE IV
CONTROLLED SAME-EVALUATOR COMPARISON BETWEEN ORIGINAL RADIO FEATURES EXTRACTED FROM RGB FRAMES AND GAUSSFM RENDERED FEATURES. BOTH USE THE SAME FROZEN SIGLIP2 HEAD, SCENE-CATEGORY SCORING, AND LABEL-FREE THRESHOLD-0.60 PLUS PEAK-CONNECTED-COMPONENT MASK CONVERSION. THIS PCC-ONLY COMPARISON IS SEPARATE FROM THE FINAL 2D OVS ROW IN TABLE I.

Scene	RADIO RGB		GaussFM	
	LocAcc	mIoU	LocAcc	mIoU
Figurines	0.7500	0.4031	0.8214	0.5134
Ramen	0.9014	0.5241	0.9014	0.6249
Teatime	0.8475	0.5429	0.8983	0.6177
Waldo Kitchen	0.7273	0.3688	0.8182	0.5268
Macro	0.8065	0.4597	0.8598	0.5707

E. Core Component Ablation

Table V isolates the main architectural choices on LERF-OVS with a controlled seed-7 protocol. Unlike Table I, which records the frozen paper-facing evaluation route, this table keeps the same seed route for the full model and its ablations. The goal is to verify whether each component contributes under the same evaluator: the contextual Gaussian field tests

TABLE V
CONTROLLED LERF-OVS COMPONENT ABLATION ON SEED 7. THE TABLE REPORTS RENDERED-FEATURE LOCACC PER SCENE PLUS MACRO LOCACC AND MIOU.

Variant	Fig.	Ramen	Tea.	Waldo	LocAcc	mIoU
Full GaussFM	0.768	0.901	0.898	0.864	0.858	0.485
w/o FGC warm-start	0.696	0.845	0.847	0.818	0.802	0.424
w/o VFA	0.768	0.859	0.915	0.818	0.840	0.480
w/o HGCF	0.768	0.873	0.898	0.818	0.839	0.507
w/o CTR	0.268	0.648	0.661	0.545	0.531	0.260

compact feature storage, foundation-space reconstruction tests the nonlinear compact feature bottleneck, view-conditioned calibration tests peak and boundary correction, and frozen-head geometry warm-start tests the geometry-aware training stage.

The largest dependency is the CTR codec. Replacing CTR with a direct 1×1 projection drops macro LocAcc from 0.858 to 0.531 and macro mIoU from 0.485 to 0.260, with every scene degraded. FGC warm-start is the next strongest component, improving macro LocAcc by 0.056 and macro mIoU by 0.061 over the feature-only run. VFA gives a smaller but consistent localization gain (+0.018 macro LocAcc) with almost unchanged mIoU. The hybrid code field mainly affects peak stability: the explicit non-hybrid variant reaches higher macro mIoU (0.507) but lower macro LocAcc (0.839), improving region overlap on Figurines/Teatime while losing peak accuracy on Ramen/Waldo. This explains why some enhancement branches can improve mIoU without preserving LocAcc.

Table VI consolidates the main quantitative ablation evidence across tasks. The largest gains come from making primitive features queryable through multiview-registration-to-field support and from the foundation-space reconstruction codec. Mask and support calibrations such as peak-component filtering, feature-only SAM3 boundary refinement, and direct-3D component support are still useful, but their deltas are smaller than the core feature-quality modules. The supplementary evidence records additional variants such as frozen official SAM3 box diagnostics and same-protocol follow-up runs not used in the main tables.

F. Storage Footprint

Table VII closes the compactness claim with explicit footprint accounting. Direct storage assumes one 1280-D fp16 RADIO feature per Gaussian and excludes ordinary RGB/geometry attributes. The latent payload is the scale-dependent semantic memory stored per Gaussian, while the feature package additionally counts scene-global heads, CTR/HCD, and VFA/refiner tensors whose fixed overhead is amortized as scene scale grows. The full checkpoint is conservative: it also includes carried Gaussian geometry/RGB tensors. Even with this accounting, the compact representation uses less storage than direct per-Gaussian RADIO features on all four LERF scenes, with larger savings on scenes with more Gaussians. Multiview Primitive Registration itself is a training and analysis bridge and does not add trained persistent parameters; if one materializes registered 1536-D SigLIP2 primitive embeddings and per-query voxel scores as a cache, that cache is reported separately and is larger than direct 1280-D fp16 feature storage. We therefore use multiview registration as streamed evidence in the main compact-storage claim rather than counting it as part of the stored scene representation. This accounting is intentionally split into three levels. The latent payload is the clean per-Gaussian feature-memory cost and gives a constant $20\times$ compression over explicit 1280-D fp16 storage. The feature-memory package is the practical deployable cost of the semantic memory: global decoders are fixed overhead and become less important as scenes grow. The full checkpoint is a conservative engineering number and should not be used alone to judge feature compression, because it also includes ordinary 3DGS geometry and RGB state that are not part of the proposed foundation-feature memory. The compression/downstream correlation audit is kept in the supplementary material: checkpoint saving ratio has only small positive correlation with rendered mIoU and negative correlation with Direct3D mIoU, so the paper does not overclaim compression alone as the source of better querying.

G. Adaptor Downstream Probes

Adaptor-consistency sweeps are kept in the supplementary material to preserve main-paper space. They show a consistent localization/region tradeoff: DINO relation and SAM3 region losses can improve thresholded overlap on Ramen, Teatime, and Figurines, but they sometimes move the single heatmap peak outside small LERF polygons. The paper therefore keeps the calibrated LERF query protocol as the main grounding result and reports the prompt-constrained downstream probes below as the compact evidence for DINO/SAM feature usability.

The broader unconstrained variants and cross-view DINO controls remain in the supplementary material. They are useful for understanding outliers, but the prompt-constrained tasks better match how frozen SAM/DINO heads are used downstream.

Table VIII separates two claims. First, rendered features improve prompt-constrained SAM3-adaptor mask mIoU over frame-wise RADIO for point prompts, box prompts, and mask propagation, while matching the frame-wise point-prompt localization. Second, using the same rendered field for DINO support and SAM-adaptor boundary refinement makes DINO propagation positive under the same evaluator: rendered features reach 0.7872/0.4677 LocAcc/mIoU versus 0.7660/0.4606 for frame-wise RADIO under the same multi-head support calibration. The higher rendered dense-matching similarity score suggests that the feature field can become over-smooth and select visually similar nearby regions. This motivates the cross-view DINO and text-heatmap protected training variants evaluated next. We also evaluate a mutual-correspondence plus homography-RANSAC control for qualitative DINO matching; it removes many outlier matches and raises rendered dense-match similarity to 0.9277, but it does not improve the same robust DINO only mask-propagation mIoU beyond

TABLE VI
UNIFIED QUANTITATIVE CONTRIBUTION RANKING. EACH ROW COMPARES A REFERENCE AND AN ENABLED VARIANT UNDER THE SAME LOCAL PROTOCOL UNLESS MARKED DIAGNOSTIC. SCORE IS THE MEAN POSITIVE DELTA OVER THE LISTED METRICS; SECONDARY REGRESSIONS ARE KEPT IN THE METRIC-DELTA COLUMN.

Contribution	Task	Reference → Variant	Metric deltas	Score	Status
VPR-to-field primitive feature registration	LERF Direct3D	raw Gaussian-center score -¿ VPR-to-field compact primitive score	mIoU: 0.0460-¿0.4120 (+0.3660); Acc@0.25: 0.0710-¿0.5880 (+0.5170)	0.442	core direct-field evidence
Foundation-space reconstruction codec	LERF rendered architecture	w/o CTR -¿ full GaussFM seed-7 model	LocAcc: 0.5310-¿0.8580 (+0.3270); mIoU: 0.2600-¿0.4850 (+0.2250)	0.276	core architecture
Final rendered-view readout vs frame-wise RADIO	2D frame-wise-RADIO-vs-field feature usability	frame-wise RADIO reference -¿ GaussFM rendered field + feature-only boundary readout	LocAcc: 0.7985-¿0.8598 (+0.0613); mIoU: 0.4634-¿0.5889 (+0.1255)	0.093	main claim evidence
FGC/FDH geometry-aware warm-start	LERF rendered architecture	w/o FGC warm-start -¿ full GaussFM seed-7 model	LocAcc: 0.8020-¿0.8580 (+0.0560); mIoU: 0.4240-¿0.4850 (+0.0610)	0.058	core architecture
Direct3D prompt ensemble + component support policy	LERF Direct3D	strict single-prompt pure one-map -¿ score-component guarded compact readout	mIoU: 0.4489-¿0.5014 (+0.0525); Acc@0.25: 0.6724-¿0.7044 (+0.0321); Boundary-F: 0.6124-¿0.6305 (+0.0181)	0.034	main compact direct readout
DINOv3 dense matching feature readout	2D frozen-head downstream	frame-wise RADIO reference -¿ GaussFM rendered field	Mean score: 0.8547-¿0.9048 (+0.0501); HitRate: 0.5723-¿0.5396 (-0.0327)	0.025	downstream readout evidence with caveat
SAM3 point-prompt feature readout	2D frozen-head downstream	frame-wise RADIO reference -¿ GaussFM rendered field	mIoU: 0.3700-¿0.4173 (+0.0473); LocAcc: 1.0000-¿1.0000 (+0.0000)	0.024	downstream readout evidence
Peak-component rendered mask readout	LERF rendered-view grounding	threshold 0.60 mask -¿ threshold 0.60 + peak component	mIoU: 0.5243-¿0.5707 (+0.0464); LocAcc: 0.8712-¿0.8598 (-0.0114)	0.023	readout policy

TABLE VII
STORAGE FOOTPRINT ACCOUNTING. DIRECT STORES ONE 1280-D FP16 RADIO FEATURE PER GAUSSIAN. LATENT IS THE COMPACT PER-GAUSSIAN SEMANTIC PAYLOAD. FEATURE PACKAGE ADDS SCENE-GLOBAL HEADS, CTR/HCD, AND VFA/REFINER TENSORS; THESE FIXED TENSORS ARE AMORTIZED AS SCENE SCALE GROWS. FULL CHECKPOINT ADDITIONALLY COUNTS CARRIED 3DGS GEOMETRY/RGB STATE. LOWER IS BETTER FOR STORAGE; HIGHER SAVING RATIOS ARE BETTER.

Scene	#G	Direct MiB	Latent MiB	Save	Feature pkg. MiB	Save	Full ckpt. MiB	Save
Figurines	168,791	412.1	20.6	20.0×	199.0	2.07×	237.0	1.74×
Ramen	382,687	934.3	46.7	20.0×	225.1	4.15×	311.2	3.00×
Teatime	460,157	1123.4	56.2	20.0×	234.5	4.79×	338.1	3.32×
Waldo Kitchen	691,728	1688.8	84.4	20.0×	262.8	6.43×	418.5	4.04×

TABLE VIII
PROMPTABLE SAM3-ADAPTOR AND DENSE DINOV3-ADAPTOR TASKS ON LERF-OVS. THESE PROBES USE THE SAME ANNOTATIONS AS THE GROUNDING BENCHMARK BUT REPLACE TEXT QUERIES WITH ADAPTOR-SPACE PROMPTS OR SOURCE-TARGET CORRESPONDENCES. SAM3 POINT/BOX PROMPTS ARE PROMPT-CONSTRAINED, MASK PROPAGATION USES SOURCE-VIEW SUPPORT ONLY, AND NO EXTERNAL SAM3 MASK DECODER IS CALLED. DINO MASK PROPAGATION USES SOURCE-BACKGROUND CONTRAST 1.10 PLUS LABEL-FREE MULTI-HEAD SUPPORT CALIBRATION: TOP-k FOREGROUND POOLING, 2.0× PROPAGATED-AREA SCALING, PEAK-COMPONENT CLEANUP, AND SAM-ADAPTOR BOUNDARY REFINEMENT.

Task	Frame-wise RADIO		Rendered GaussFM	
	LocAcc/Hit	mIoU/Score	LocAcc/Hit	mIoU/Score
SAM3 point prompt	1.0000	0.3700	1.0000	0.4173
SAM3 box prompt	0.8702	0.6560	0.8221	0.6638
SAM3 mask propagation	0.7872	0.3583	0.6596	0.3756
DINOv3 dense matching	0.5723	0.8547	0.5396	0.9048
DINOv3 mask propagation + SAM-boundary calibration	0.7660	0.4606	0.7872	0.4677

frame-wise RADIO. We therefore keep RANSAC as a visualization control and use the multi-head DINO-support/SAM-boundary calibration for the quantitative frozen-head table.

H. ScanNet Direct Point-Query Transfer

Table IX evaluates cross-domain feature usability on the VALA-aligned ScanNet-8 scene subset: `scene0000_00`, `scene0062_00`, `scene0070_00`, `scene0097_00`, `scene0140_00`, `scene0347_00`, `scene0400_00`, and `scene0590_00`. This follows a public VALA-aligned direct point-query setting with every-20-frame ScanNet training splits and mIoU/mAcc evaluation on GT semantic point clouds. This is not presented as a full ScanNet semantic segmentation leaderboard result; it is a split-aligned direct probe of the learned feature field. The GaussFM result uses the cross-view DINO-consistent compact field with a label-free contextual kNN point-query classifier ($k = 16$, 80 candidate Gaussians), single-template text prompts, and scene-mean logit calibration at $\alpha = 0.45$. A label-free spatial logit propagation step averages class logits over 12 nearest 3D neighbours before the final argmax. It reaches 36.55/50.57, 42.78/72.85, and 57.85/77.93 mIoU/mAcc on the 19/15/10-class splits. The baseline rows follow the same reproduced VALA-aligned point-query protocol. A diagnostic proposal-memory variant improves some fine-grained 19-class categories but lowers the final 15-/10-class balance, so it is kept as an auxiliary ScanNet control in the supplementary material.

Table X summarizes the main ScanNet failure modes for the GaussFM result. The weakest classes are either small decorative regions such as *picture*, broad cluttered surfaces such as *table*, or classes with sparse scene support such as *toilet*; unstable classes such as *door*, *window*, and *refrigerator* vary strongly with view coverage and object extent.

I. Efficiency and Cost

Table XI reports lower-is-better single-query latency derived from frozen profile artifacts, separate from storage footprint accounting. The reported units match each protocol: rendered-view LERF uses view-query latency, direct 3D LERF uses object-query latency after query-select-render mask generation, and ScanNet uses class-query latency. These values are

TABLE IX
QUANTITATIVE EVALUATION OF SCANNET-V2 OPEN-VOCABULARY POINT QUERYING UNDER A VALA-ALIGNED EIGHT-SCENE PROTOCOL. ALL ROWS FOLLOW THE SAME CLASS SPLITS AND REPRODUCED POINT-QUERY EVALUATOR; RESULTS ARE REPORTED IN PERCENT.

Method	19 mIoU	19 mAcc	15 mIoU	15 mAcc	10 mIoU	10 mAcc
LangSplat	2.45	8.59	3.45	13.21	6.48	21.89
LangSplatV2	14.75	25.47	17.09	35.68	22.83	41.52
OpenGaussian	27.73	42.01	29.67	46.15	39.93	57.34
Dr. Splat	29.31	47.68	33.25	54.33	44.19	65.19
OccamLGS	31.93	48.93	34.25	53.71	45.16	64.39
VALA	32.11	50.05	35.10	54.77	46.21	65.61
GaussFM	**36.55**	**50.57**	**42.78**	**72.85**	**57.85**	**77.93**

TABLE X
SCANNET VALA8 CATEGORY STABILITY FOR THE GAUSSFM DINO-CV CONTEXTUAL KNN CLASSIFIER WITH SPATIAL LOGIT PROPAGATION. STATISTICS ARE PER-CLASS IOU ACROSS SCENES WHERE THE CLASS APPEARS; "RANGE" IS MIN–MAX IOU.

Split	Weakest class	Scenes	Mean IoU	Range	Most unstable class	Std IoU	Range
19	picture	3	0.018	0.000–0.053	refrigerator	0.297	0.004–0.827
15	table	4	0.126	0.000–0.374	door	0.238	0.015–0.786
10	toilet	2	0.216	0.114–0.318	window	0.244	0.180–0.867

TABLE XI
SINGLE-QUERY LATENCY EVIDENCE. LOWER IS BETTER. VALUES ARE DERIVED FROM FROZEN PROFILE ARTIFACTS AND ARE CONSERVATIVE BECAUSE THE PROFILED COMMANDS INCLUDE EVALUATOR OVERHEAD AND, FOR LERF RENDERED-VIEW RUNS, VISUALIZATION/RADIO-TARGET BRANCHES. DIRECT 3D LATENCY INCLUDES QUERY-SELECT-RENDER MASK GENERATION.

Task	Unit	Queries	Latency ms/query	Peak VRAM MiB
LERF rendered-view OVS	view-query	356	350.5	2076
LERF direct 3D OVS	object-query	208	2367.7	–
ScanNet point query	class-query	389	387.9	1666

conservative because the frozen profiles include evaluator work and, for rendered-view LERF, visualization/RADIO-target branches; clean warm-GPU microbenchmarks are kept out of the main claim until run under a single instrumentation harness. Training throughput remains supplementary log-derived evidence because training wall time and explicit GPU telemetry are different measurement types. The important point is therefore not to present these numbers as optimized system latency, but to avoid conflating scene-level batch evaluation time with per-query deployment cost. The direct-3D row is expectedly slower because it includes selected-primitive rendering for mask evaluation; primitive scoring itself is the compact-memory operation. We therefore interpret the latency table as conservative protocol-level evidence and separate it from the storage claim.

J. Qualitative Results

The main qualitative comparison in Fig. 3 follows the LERF-OVS taxonomy used in recent open-vocabulary Gaussian-field work: rendered 2D querying and direct 3D primitive querying are shown side by side on the same annotated view and text query. The 2D prior panels use our local LangSplat V2 reproduction, while the 3D prior panels use our local Dr. Splat reproduction. The GaussFM 3D panels use the deployed compact field with the fixed support-calibrated primitive selection rule; they do not read a multiview-registration feature cache and do not call the official RGB SAM3 mask decoder at evaluation time.

Fig. 4 gives the corresponding direct point-query visualization on ScanNet. We use binary class-query point clouds rather than full semantic coloring, because the evaluation protocol asks whether a frozen text query retrieves a target class under the VALA-aligned split.

Fig. 5 visualizes the direct-3D support-calibration ablation on small or fragmented LERF queries. The base compact selector uses primitive scores from the compact field, while the final selector adds the prompt ensemble and label-free component-support calibration used by the deployed compact direct query.

Fig. 6 isolates the rendered-view boundary calibration used by the feature-only SAM3-adaptor mask head. Unlike the frozen official SAM3 diagnostic, this head is conditioned on reconstructed GaussFM RADIO-compatible features and prompt embeddings; it does not call the official RGB SAM3 decoder at evaluation time.

Fig. 7 compares reconstructed scene features with frame-wise RADIO features under the frozen downstream probes used in Table VIII. These examples visualize the same claim as the quantitative probe table: the compact scene field is not merely a compressed copy of a single view, but a multiview feature memory that can produce more stable downstream responses.

Additional rendered heatmap examples, alpha/depth boundary cases, and extended SAM/DINO-adaptor controls are kept in the supplementary material so the main paper focuses on one representative figure per claim.

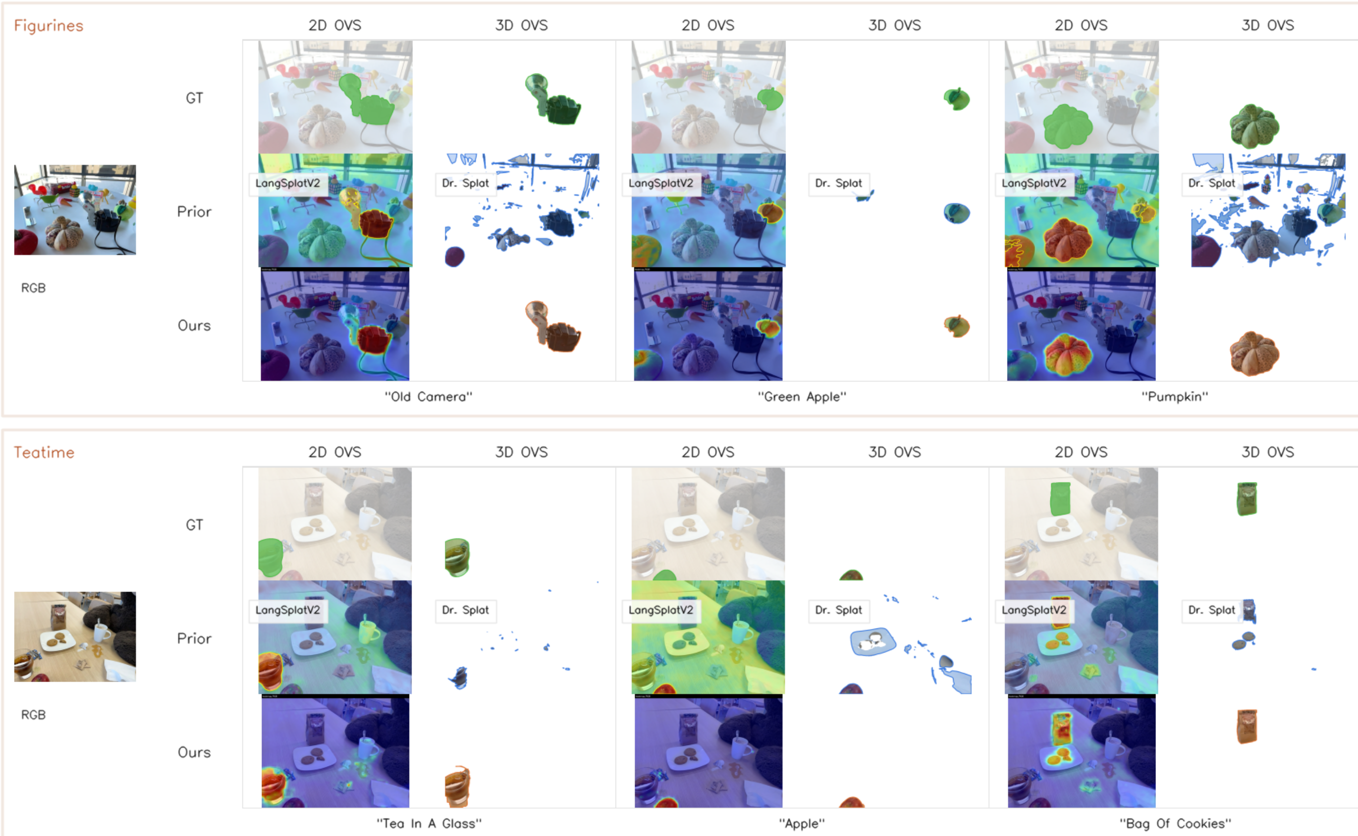

Fig. 3. Qualitative comparison of 2D and 3D open-vocabulary querying on LERF-OVS. For each text query, the 2D columns visualize rendered-view mask/heatmap outputs on RGB, while the 3D columns visualize direct Gaussian-level selection rendered on a white background. The prior result uses reproduced local baselines: LangSplatV2 for 2D OVS and Dr. Splat for 3D OVS. GaussFM uses one compact foundation-feature Gaussian map for both query modes; the 3D panels do not use a multiview-registration cache or official RGB SAM3 mask decoder.

TABLE XII
REPRESENTATIVE FRAGILE LERF-OVS CATEGORIES FROM THE FROZEN RENDERED-FEATURE EVALUATION. LOW LOCACC WITH NONZERO MIOU HIGHLIGHTS THE PEAK-VS-REGION TRADE-OFF.

Scene	Category	LocAcc	mIoU
Figurines	bag	0.00	0.00
Figurines	miffy	0.00	0.01
Teatime	dall-e brand	0.00	0.01
Figurines	jake	0.00	0.02
Waldo Kitchen	ketchup	0.00	0.03
Waldo Kitchen	spatula	0.00	0.03
Ramen	corn	0.00	0.10
Figurines	red toy chair	0.00	0.13

K. Failure Analysis

Table XII turns the qualitative small-object observation into a quantitative failure analysis. The lowest-ranked categories are mostly tiny or visually ambiguous objects, such as Figurines bags, character toys, and isolated kitchen tools. The failure pattern is not only low region overlap: some categories show nonzero mIoU but zero or partial LocAcc, which means the rendered heatmap covers a plausible object region while the single maximum lands outside the annotated polygon. This is why the paper treats mIoU improvements from adaptor or cross-view losses conservatively unless localization accuracy is preserved.

V. DISCUSSION

a) Why feature reconstruction matters.: The LERF-OVS results show that rendered RADIO-like features retain enough semantic structure for open-vocabulary localization. The ScanNet results suggest that the same representation can be probed outside the LERF object-query setting. The LERF direct-selection experiment further clarifies the boundary of this claim: the compact field supports direct primitive querying, Multiview Primitive Registration provides an auditable training bridge for primitive evidence, and support-calibrated selection is needed to convert primitive scores into stable object masks. Bare score-threshold selection is still weaker than object-aware grouping on fragmented scenes, so we distinguish primitive scoring quality from boundary and support calibration quality.

b) Baseline scope.: The main LERF and ScanNet quantitative tables use the reproduced protocols reported in this paper. Historical cross-paper provenance tables are retained only as auxiliary audit material and are not the basis for the main ranking claims.

c) Why the compact field can outperform frame-wise RADIO.: GaussFM is trained from frame-wise RADIO features, but its rendered query path is not restricted to a single frame. Multiview reconstruction, visibility-aware regularization, and support-calibrated selection can suppress view-specific noise

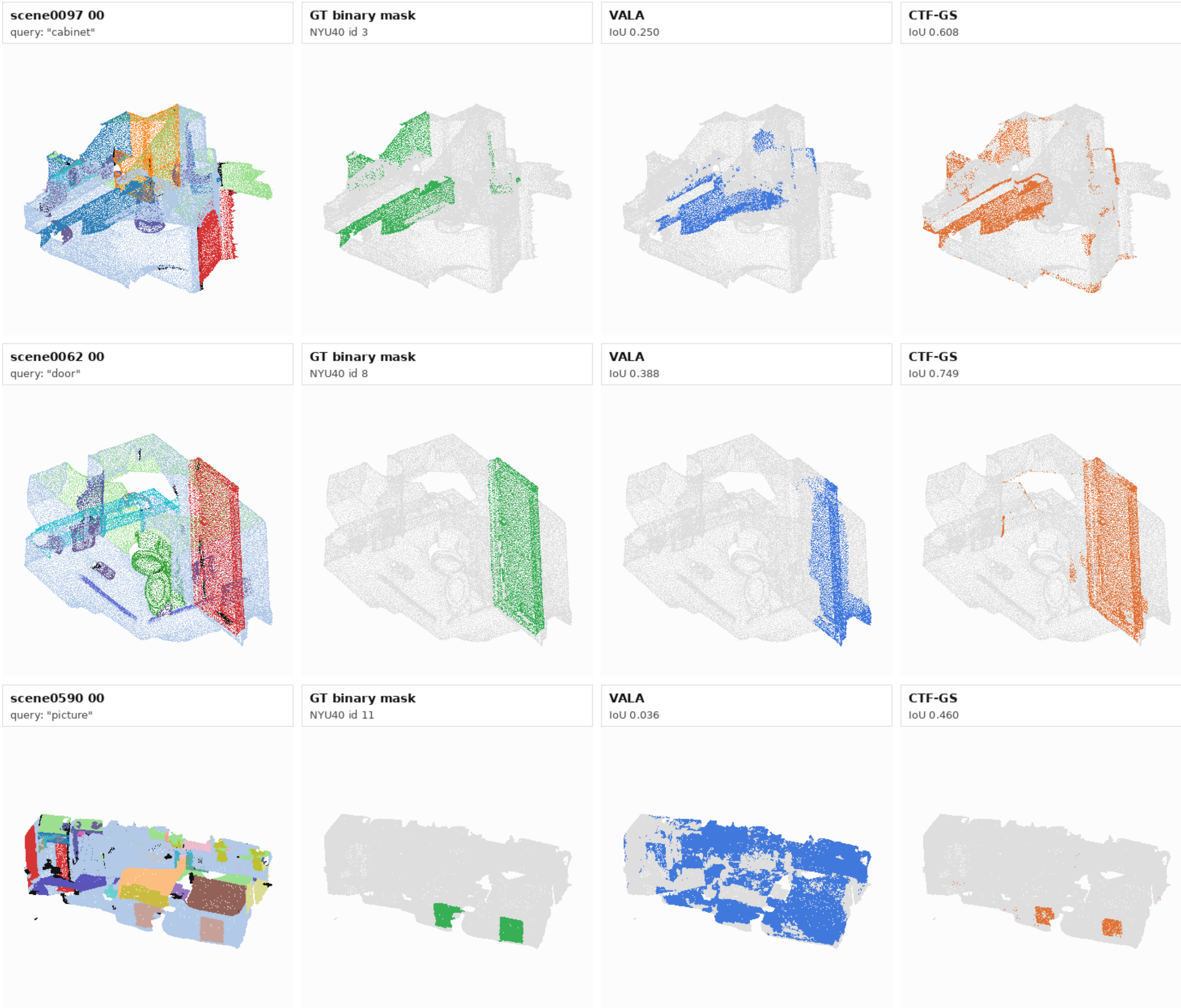

Fig. 4. ScanNet open-vocabulary 3D query qualitative comparison. Each row shows a binary text-class query on a VALA-aligned ScanNet scene: scene overview, GT class points, our local VALA compatibility reproduction, and GaussFM direct point-query prediction. Gray points are non-query classes.

and recover a scene-consistent response. This explains why the rendered field can outperform frame-wise RADIO under the same frozen SigLIP2, SAM3-adaptor, and DINOv3 probe tasks, while still being limited by frame-wise feature resolution and annotation ambiguity on small objects.

VI. Limitations

GaussFM depends on a pretrained Gaussian geometry backbone and does not claim to improve RGB reconstruction. Small objects remain challenging because RADIO feature resolution, Gaussian support density, and rendered feature alignment limit heatmap sharpness. The ScanNet protocol is a direct point-query feature probe, not a replacement for a full standardized semantic segmentation benchmark. The main LERF and ScanNet tables use the reproduced protocols reported in this paper; historical cross-paper provenance notes are kept only for auditability. The deployed compact direct-3D result uses GT-free color-edge and score-component support calibration. This does not invoke a learned RGB segmentation network or official RGB SAM decoder, but it should be understood as a lightweight support-calibration step rather than a completely image-free selector. The frozen official SAM3 box result is therefore kept as a post-selection diagnostic rather than the core compact-memory claim. The storage table reports feature footprint separately from full checkpoint footprint; a deployment package should keep the same separation between persistent feature parameters, transient rendering buffers, and optional downstream heads.

VII. Conclusion

GaussFM turns a 3D Gaussian scene into a reusable compact foundation-feature memory. Its central claim is deliberately narrower than generic semantic Gaussian learning: a low-dimensional contextual Gaussian field can reconstruct RADIO-

Fig. 5. Qualitative ablation of compact direct-3D support calibration. The selector recovers small or fragmented object support without a multiview-registration feature cache or official RGB SAM3 mask decoder; rendering is used only to visualize and evaluate the selected 3D primitives.

compatible scene features, absorb sparse multiview primitive evidence, and support rendered-view, direct-primitive, and point-level open-vocabulary queries without deploying a high-dimensional multiview feature cache. The LERF-OVS, direct-3D, ScanNet, frame-wise RADIO comparison, storage, latency, and ablation results jointly support this claim. The remaining limitations are also clear: small fragmented objects and boundary-sensitive queries still depend on support calibration, and the ScanNet protocol should be interpreted as direct foundation-feature probing rather than full semantic-segmentation replacement. These boundaries make the result reproducible and falsifiable while preserving the main contribution: compact reconstructive foundation features can be made a first-class 3D scene representation.

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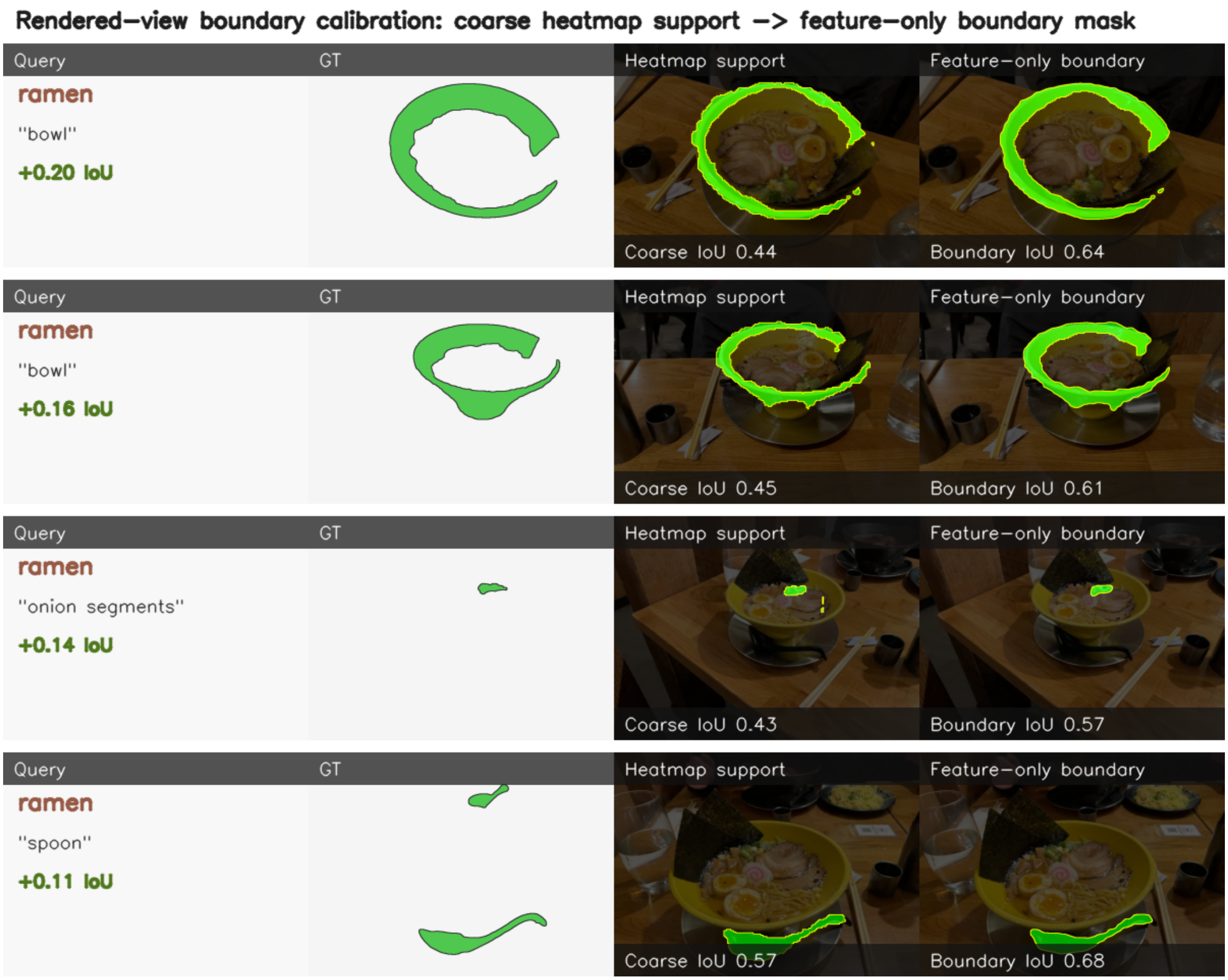

Fig. 6. Rendered-view boundary calibration qualitative examples. The coarse heatmap support is converted into a feature-only boundary mask using the prompt-conditioned SAM3-adaptor head. The examples show per-query IoU gains without invoking an external RGB SAM3 decoder.

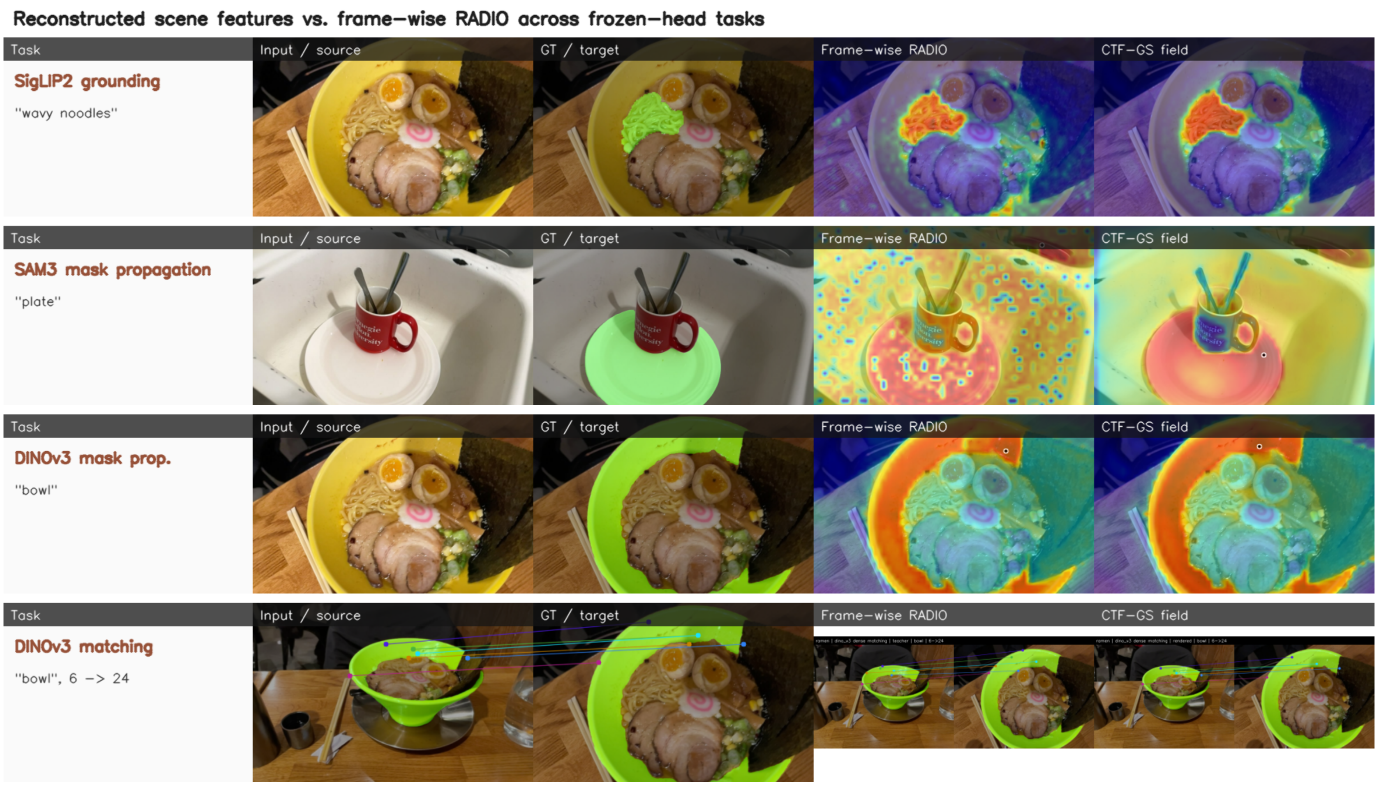

Fig. 7. Reconstructed scene features versus frame-wise RADIO under frozen-head tasks. Columns compare the task input/GT, frame-wise RADIO response, and GaussFM reconstructed field response for SigLIP2 grounding, SAM3-adaptor mask propagation, DINOv3 mask propagation, and DINOv3 dense matching.